

# AVIRAL SACHDEVA

B.Tech Computer Science & Engineering | AI & Machine Learning

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India

## EDUCATION

GLA University

Expected 2029

Bachelor of Technology in Computer Science and Engineering

## TECHNICAL SKILLS

**AI & Machine Learning:** Computer Vision, Object Detection, Reinforcement Learning, Generative AI (Gemini API)

**Data Science & Logic:** Data Pattern Analysis, Geospatial Algorithms (Haversine), Simulation Environments

**Programming Languages:** Python (Primary for AI/ML), C/C++

**Tools & Technologies:** Firebase, REST APIs, Linux Environments, Git/GitHub

## AI & DATA SCIENCE PROJECTS

**OpenEnv (Simulation-Based AI Environment)**

Reinforcement Learning

**Technologies:** Python, RL Frameworks, Simulation Logic

- Architected a training environment specifically tailored for simulation-based AI agent development.
- Designed and implemented reinforcement learning structures to allow efficient trial-and-error training protocols for autonomous decision-making models.

**Orange Pigeon (Computer Vision Defense System)**

Computer Vision

**Technologies:** Python, Object Detection Models, Edge Computing (Raspberry Pi)

- Deployed and fine-tuned a specialized computer vision model strictly trained for high-accuracy pigeon detection, minimizing false positives.
- Optimized the AI model's inference to run smoothly on edge hardware, utilizing real-time video processing.
- Developed automated Python logic to process the model's outputs and trigger physical deterrence mechanisms immediately upon detection.

**AgroVON 2.0 (AI Diagnostic Platform)**

Generative AI Integration

**Technologies:** Gemini AI API, Data Pipelines, Python

- Engineered an AI-powered diagnostic platform that analyzes incoming environmental data patterns (soil and crop metrics).

- Integrated the Gemini AI API to automate crop disease detection, turning raw sensor datasets into actionable, real-time insights for agricultural management.

## **BloodLink Pro (Algorithmic Routing Network)**

Algorithmic Optimization

**Technologies: Geospatial Mathematics, Firebase, Python**

- Designed a real-time medical matching network driven by complex geospatial data calculations.
- Implemented the Haversine formula to process coordinate data, optimizing the backend to match blood donors and receivers based on strict proximity constraints.

## **HACKATHONS & LEADERSHIP**

- ▷ **Devcation Delhi 2026 (IIT Delhi):** Competed in the AI-focused Duality Track. Focused on building and deploying the OpenEnv simulation-based reinforcement learning project under stringent time constraints.
- ▷ **AI Summit:** Led the team as **Team Lead**. Delegated AI model training tasks, managed the integration pipeline, and successfully presented the AI/ML prototypes to the judging panel.
- ▷ **Project Exhibition:** Served as **Team Lead** for a major institutional exhibition, presenting complex data flows and AI workflows to an audience of tech professionals and faculty.
- ▷ **Smart Krishi Hackathon:** Showcased the AgroVON 2.0 system, focusing heavily on how AI diagnostics and data analysis can solve scaleable agricultural challenges.